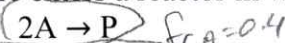


SPECIAL NOTE: You must show all of your work to receive full credit. Do all calculations in the exam booklet provided. To receive full credit, write your final answer in the spaces provide on this sheet. You must turn this sheet in with your exam booklet.

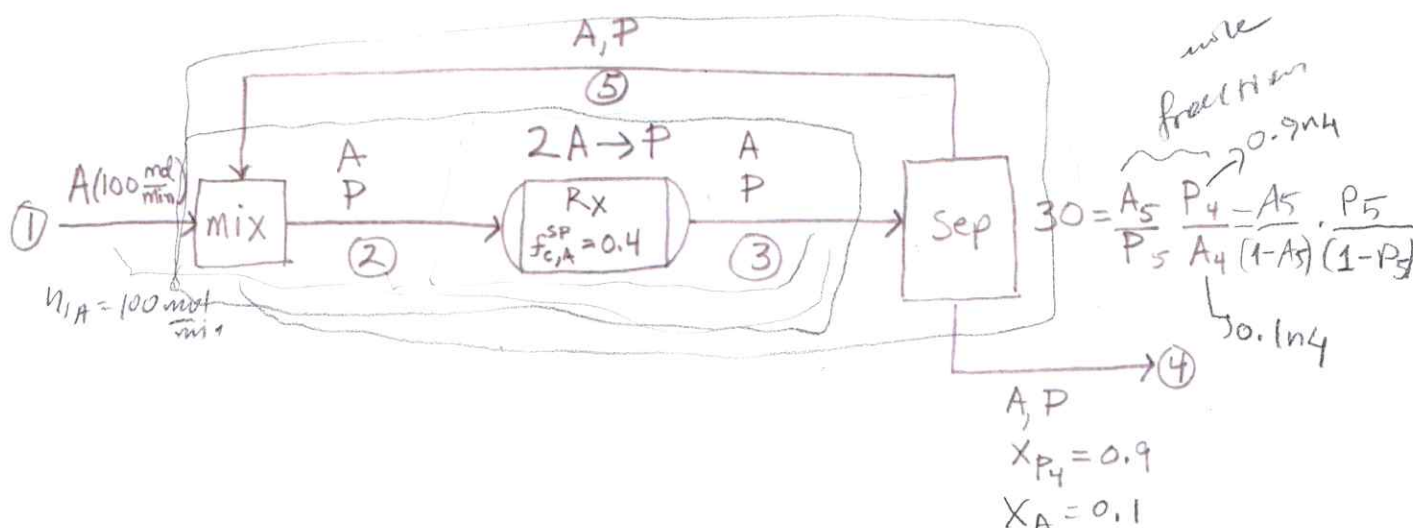
- (1) (50 pts.) Consider the block flow diagram below. Notice that the streams are labeled 1 through 5. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains only species A. This stream is mixed with a recycle stream containing A and P where P is the reaction product. The mixture enters a reactor in which one reaction takes place:



The reactor converts 40% of A entering the reactor. The outlet stream from the reactor goes to a separator. Both streams leaving the separator are in equilibrium containing some A and P such that:

$$\alpha_{AB} = \frac{A_5 P_4}{P_5 A_4}$$

where the separation factor $\alpha_{AB} = 30$. Stream 5 is the stream enriched in A and is recycled back to the reactor. Stream 4 is enriched in the product P such that it is 90 mol% P. The fresh feed has species flow rates of $A_1 = 100$ mol A/min.



- (a) Find all of the species molar flow rates not given in the problem statement. Fill in the table below with the appropriate species flow rates in mole/min to receive full credit. (45 pts.)

Species/Stream →	1	2	3	4	5
A	100				
P	----				

- (b) Calculate the overall fractional conversion based on species A (5 pts.)

$f_A^{ov} = \frac{n_1 - n_{A3}}{n_1}$ *Excess*

$f_{c,A}^{ov} = \underline{\hspace{2cm}}$

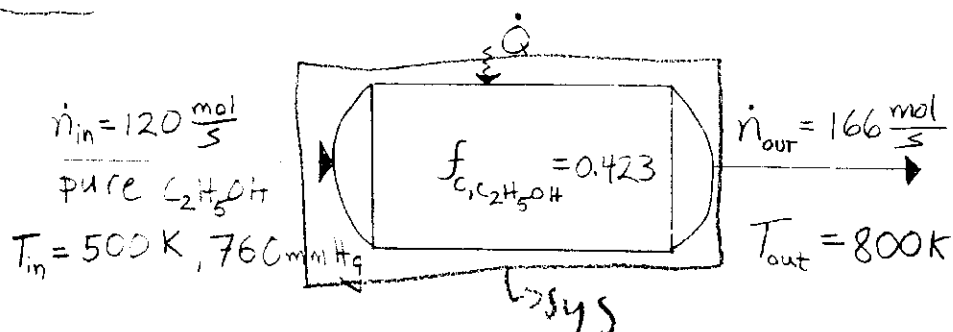
(2) (50 pts.) Acetaldehyde (CH_3CHO) is produced by dehydrogenation of ethanol ($\text{C}_2\text{H}_5\text{OH}$).



An undesired side reaction produces ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$)



This process is carried out in a single reactor operating at steady state. The reactor has a feed rate of 120 mol/s of pure ethanol at 500K and 760 mm Hg. The reactor total product flow rate is 166 mol/s at 800K. The reactor converts 42.3% of the ethanol.



Data: The heat capacity in $\text{J/mol}^\circ\text{C}$ is given by $C_p = A + Bt$ where $t = T(\text{K})/1000$. The table below gives the constants A and B . $C_p dT = (A + BT) dT$

Species	A	B
$\text{C}_2\text{H}_5\text{OH}$	32.3	120
CH_3CHO	30.7	88.2
$\text{CH}_3\text{COOC}_2\text{H}_5$	60.9	191
H_2	28.4	1.54

$$t = \frac{T(\text{K})}{1000}$$

$$dT = \frac{1}{1000} dt$$

- (a) Calculate the all the species molar flow rates for the product stream in mol/s and write them into the Input/Output Table below. Also, calculate the two extents of reaction, ξ_1 and ξ_2 . (20 pts.)

$$\xi_1 = \underline{41.24} \text{ mol/s} \quad \xi_2 = \underline{4.76} \text{ mol/s}$$

- (b) Finish filling enthalpy values into the Input/Output Table below. Use a reference state of $T_{\text{ref}} = 25^\circ\text{C}$. (10 pts.)

Species	$\dot{n}_{in}$ (mol/s)	$\bar{H}_{in}$ (kJ/mol)	$\dot{n}_{out}$ (mol/s)	$\bar{H}_{out}$ (kJ/mol)
$\text{C}_2\text{H}_5\text{OH}$	120	-3.0567	69.24	16.633
CH_3CHO	---	---	41.24	12.229
$\text{CH}_3\text{COOC}_2\text{H}_5$	---	---	4.76	26.478
H_2	---	---	50.76	0.2275

- (c) Calculate the heat rate in kJ/s for the reactor. Indicate clearly whether it is heat in or out of the reactor. (20 pts.)

$$\dot{Q} = \underline{5116.538} \text{ kJ/s}$$

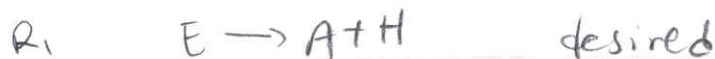
I.D. #	
Name	Katerina Bompai
Subject	
Course	Section
Instructor	
Date	

Receiving or giving aid in a final examination is a cause for dismissal from the University.

(Begin Writing on this page)

$$\begin{array}{r}
 15 \\
 \hline
 2 \quad 44 \\
 \hline
 1 \quad 59
 \end{array}$$

②



Species Balances

$$\dot{n}_{E,in} = \dot{n}_{in} \quad , \quad \dot{n}_{T,out} = 166 \text{ mol/s}$$

$$166 = \dot{n}_{E,out} + \dot{n}_{A,out} + \dot{n}_{EA,out} + \dot{n}_{H,out}$$

E: $\dot{n}_{E,out} = \dot{n}_{in} - \dot{J}_1 - 2\dot{J}_2 \Rightarrow \dot{n}_{E,out} = 120 \text{ mol/s} - \dot{J}_1 - 2\dot{J}_2$

desired A: $\dot{n}_{A,out} = 0 + \dot{J}_1 \Rightarrow \boxed{\dot{n}_{A,out} = 50.76 - 2\dot{J}_2}$

undesired EA: $\dot{n}_{EA,out} = 0 + \dot{J}_2$

H: $\dot{n}_{H,out} = 0 + \dot{J}_1 + 2\dot{J}_2 = 50.76 \text{ mol/s}$

$$f_{C,E} = \frac{\dot{n}_{E,in} - \dot{n}_{E,out}}{\dot{n}_{in}} = 0.423 \Rightarrow \frac{120 - (120 - \dot{J}_1 - 2\dot{J}_2)}{120} = 0.423$$

$$0.423 = \frac{\dot{J}_1 + 2\dot{J}_2}{120} \Rightarrow \boxed{\dot{J}_1 + 2\dot{J}_2 = 50.76 \text{ mol/s}}$$

$$\Rightarrow \boxed{\dot{J}_1 = 50.76 \text{ mol/s} - 2\dot{J}_2}$$

$$\dot{n}_{E,out} = 120 - (\dot{J}_1 + 2\dot{J}_2) = 69.24 \text{ mol/s}$$

$$\dot{n}_{TOT} = \dot{n}_{E,out} + \dot{n}_{A,out} + \dot{n}_{EA,out} + \dot{n}_{H,out}$$

$$166 \text{ mol/s} = [67.24 + (50.76 - 2j_2) + j_2 + 50.76] \text{ mol/s}$$

$$166 \text{ mol/s} = (170.76 - j_2) \text{ mol/s}$$

$$j_2 = 170.76 - 166 \Rightarrow \boxed{j_2 = 4.76 \text{ mol/s}} = \dot{n}_{EA,out}$$

$$j_1 = 50.76 - 2(j_2) = \boxed{j_1 = 41.24 \text{ mol/s}} = \dot{n}_{A,out}$$

20/20

b) For EB: PE, KE & 0, $W_s = 0$

$$\begin{aligned} Q = \Delta H &= \sum_j \dot{n}_j \hat{H}_j + \sum_e (n_e \hat{H}_e) - \sum_i (n_i \hat{H}_{in}) \\ &= \sum_j \dot{n}_j \hat{H}_{r,j} + \sum_e \dot{n}_e \hat{H}_{r,e} + \sum_e (n_e \hat{H}_e) - \sum_i (n_i \hat{H}_{in}) \end{aligned}$$

1) Find ΔH_{r1} , ΔH_{r2} , $\Delta \hat{H}_r^\circ = \sum_i \nu_i \Delta \hat{H}_f^\circ(\text{products}) - \sum_j \nu_j \Delta \hat{H}_f^\circ(\text{reactants})$

$$\Delta \hat{H}_{r1}^\circ = \Delta \hat{H}_f^\circ(H) + \cancel{\Delta \hat{H}_f^\circ(H_2)}_{\text{at } 25^\circ\text{C}} - \Delta \hat{H}_f^\circ(E)$$

$$\Delta \hat{H}_{r1}^\circ = (-166.2 \frac{\text{kJ}}{\text{mol}}) - (-234.95 \frac{\text{kJ}}{\text{mol}}) = 68.75 \frac{\text{kJ}}{\text{mol}}$$

$$\begin{aligned} \Delta \hat{H}_{r2}^\circ &= \Delta \hat{H}_f^\circ(EA) + 2 \cancel{\Delta \hat{H}_f^\circ(H_2)} - 2 \Delta \hat{H}_f^\circ(E) \\ &= -444.5 - 2(-234.95) = 25.4 \frac{\text{kJ}}{\text{mol}} \end{aligned}$$

2) Find Enthalpies:

$$H = \int_{T_1}^{T_2} C_p dT$$

$$T_{in} = 500\text{ K} - 273.15 = 226.85^\circ\text{C}$$

$$T^\circ\text{C} \rightarrow \text{K}$$

$$^\circ\text{C} + 273.15 \neq$$

$$\text{K} \rightarrow ^\circ\text{C} \quad 25^\circ\text{C} + 273.15$$

$$\hat{H}_{E,in} = \frac{1}{1000} \int_{226.85^\circ\text{C}}^{25^\circ\text{C}} (A + Bt) dt$$

$$C_p \text{ in J/mol}^\circ\text{C}$$

$$\hat{H}_{E,in} = \frac{1}{1000} \left(At + \frac{Bt^2}{2} \right) \Big|_{226.85^\circ\text{C}}^{25^\circ\text{C}}$$

wrong limits

$$= \frac{1}{1000} \left(32.3(25^\circ\text{C} - 226.85) + \frac{120}{2}(25^2 - (226.85)^2) \right)$$

$$\hat{H}_{E,in} = -3056.68 \text{ J/mol} = -3.0567 \text{ kJ/mol}$$

$$\hat{H}_{E,out} = \frac{1}{1000} \int_{25^\circ\text{C}}^{526.85^\circ\text{C}} (A + Bt) dt$$

$$= \frac{1}{1000} \left(32.3(526.85 - 25) + \frac{120}{2}(526.85^2 - 25^2) \right)$$

4/10

$$\hat{H}_{E,out} = 16632.97 \text{ J/mol} \approx \hat{H}_{E,out} = 16.633 \text{ kJ/mol}$$

$$\hat{H}_{A,out} = \frac{1}{1000} \int_{25^\circ\text{C}}^{526.85^\circ\text{C}} (A + Bt) dt = \frac{1}{1000} \left(At + \frac{Bt^2}{2} \right) \Big|_{25^\circ\text{C}}^{526.85^\circ\text{C}}$$

$$\hat{H}_{A,out} = \frac{1}{1000} \left(30.7(526.85^\circ\text{C} - 25) + \frac{88.2}{2}(526.85^2 - 25^2) \right)$$

$$\hat{H}_{A,out} = 12228.72 \text{ J/mol} = 12.229 \text{ kJ/mol}$$

$$\hat{H}_{EA,out} = \frac{1}{1000} \int_{25^\circ\text{C}}^{526.85^\circ\text{C}} (A + Bt) dt = \frac{1}{1000} \left(60.9(526.85 - 25) + \frac{191}{2}(526.85^2 - 25^2) \right)$$

$$\hat{H}_{EA,out} = 26478.9 \text{ J/mol} = 26.479 \text{ kJ/mol}$$

$$\hat{H}_{H,out} = \frac{1}{1000} \left(28.4(526.85 - 25) + \frac{1.54}{2}(526.85^2 - 25^2) \right)$$

$$\hat{H}_{H,out} = 227.5 \text{ J/mol} = 0.2275 \text{ kJ/mol}$$

$$\dot{Q} = \underbrace{\sum_1 \dot{\Delta \hat{H}}_{r1} + \sum_2 \dot{\Delta \hat{H}}_{r2}}_{\text{out}} + \underbrace{\sum_c (n_c \hat{H}_c)}_{(2)} - \sum_i (n_i \hat{H}_i)$$

$$(1) \quad 41.24 \frac{\text{mol}}{\text{s}} (68.75 \frac{\text{kJ}}{\text{mol}}) + 4.76 (25.4 \frac{\text{kJ}}{\text{mol}}) = \underline{2956.154 \text{ kJ/s}}$$

$$(2) \quad \left[69.24 (16.633) + 41.24 (12.229) + 4.76 (26.479) + 50.76 (0.2275) \right] - [120 (-3.0567)]$$

$$= 1773.58082 - (-366.804) = \underline{2160.385 \text{ kJ/s}}$$

$$\dot{Q} = 2956.154 \text{ kJ/s} + 2160.385 \text{ kJ/s} = 5116.538$$

20/20

44/50

$$100 - \dot{n}_2 = \dot{n}_5$$

mixer: $\dot{n}_5$

$$\dot{n}_{A1} + \dot{n}_{A5} + \dot{n}_{P5} = \dot{n}_2$$

$$\dot{n}_{A2} + \dot{n}_{P2}$$

$$100 = \dot{n}_2 - \dot{n}_5$$

(1)

(2)



Reactor Balance:

$$\dot{n}_{A3} = \dot{n}_{A2} - \dot{J}$$

$$\dot{n}_{P3} = \dot{n}_{P2} + \dot{J}$$

$$0.4 = \dot{J}$$

$$\dot{n}_{A2} + \dot{n}_{A5}$$

$$\dot{J}_{SP} = 0.4$$

(1) →

$$100 + \dot{n}_{A5} + \dot{n}_{P5} - \dot{n}_{A2} = \dot{n}_2$$

$$100 + \dot{n}_{A5} - \dot{n}_{A2}$$

$$\dot{n}_{A3} = (\dot{n}_{A1} + \dot{n}_{A5}) - \dot{J}$$

$$\dot{n}_{P3} = (\dot{n}_{P2} + \dot{J})$$

Separator:

$$\dot{n}_3 = \dot{n}_5 + \dot{n}_4$$

no rxn

$$\dot{n}_{A3} + \dot{n}_{P3} = 0.9\dot{n}_4 + 0.1\dot{n}_4$$

mixer balance

$$\dot{J}_{SP} = \frac{\dot{n}_{A1} \text{ in} - \dot{n}_{A \text{ out}}}{\dot{n}_{A1}} = \frac{(100 + \dot{n}_{A5}) - \dot{n}_{A2}}{100} = 0.4$$

~~Rx + sep:~~

$$\dot{n}_4 =$$

$$\dot{n}_4 =$$

overall

5/15

10/30

0/5

$$30 = \frac{\dot{n}_{A5}}{\dot{n}_{P5}} \cdot \frac{0.9\dot{n}_4}{0.1\dot{n}_4}$$

15/50

~~Prob 10~~

Eg

$$Q=30 = \frac{A_5}{P_5} \cdot \frac{P_4}{A_4} = \frac{\dot{V}_{A5}}{(1-\eta_{A5})} \cdot \frac{\dot{V}_{P5}}{(1-\eta_{P5})} = 30$$